Python Data structure Practical Exam Slips final

Master of computer Application (Savitribai Phule Pune University)

Savitribai Phule University
F.Y.M.Sc. (Computer Application) Semester - I
(2023 Pattern)
Practical Examination on Python Programming and Data Structures
Couse Code: CA 505 MJP
Course Title: Practical course based on CA 502 MJ

Duration: 3 hours **Max Marks: 35**

Q1. Write a python program to calculate the average of numbers in a given list. [15 marks]

Q2. Create Linear Queue - Static implementation of linear Queue to perform following operations: Init, enqueue, dequeue. [15 marks]

Q3.Viva [5 marks]

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Q1. Write a program which accepts 6 integer values and prints “DUPLICATES” if any of the values entered are duplicates otherwise it prints “ALL UNIQUE”.. [15 marks] Q2. Create Priority Queue

1. Implementation of priority queue [15 Marks]

[5 Marks]

Q3.Viva

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Q1. Write a python program to display the following pattern [15 marks]

1

2 3

4 5 6

7 8 9 10

Q2. Create Searching Algorithms - Implementation of searching algorithms to search an element using: Linear Search. [15 Marks]

Q 3. Viva [5 Marks]

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Q1. .Reverse the following tuple.

aTup=(10,20,30,40,50)

[15 marks]

Q2. Sorting Algorithms - Implementation of sorting algorithms: [15 Marks]
Merge Sort.

Q3. Viva [5 Marks]

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Q1. Write a python program to create a list of tuples with the first element as the number and second element as the square of the number. [15 Marks]

[15 Marks]

Q2. Dynamic implementation of Singly Linked List to perform following operations: Create, Insert, Delete, and Display.

Q3.Viva [5 Marks]

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Q1. .Copy element 44 and 55 from the following list of tuple.
tuple1=(11,22,33,44,55,66)
. [15marks]

Q2. Create doubly Linked List - Dynamic implementation of doubly circular Linked List to perform following operations Create, Insert, Delete, Display

[15 Marks]

Q3.Viva [5 Marks]

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Q1. Write a python program to get the 5th element from front and 5th element from last of a tuple. [15 Marks]

Q2. Create Linked List Applications - Merge two sorted lists. [15 Marks]

Q3.Viva [5 Marks]

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Q1. Write a python program to find the repeated items of a tuple. [15 Marks]

Q2. Create Stack - Dynamic implementation of Stack to perform following operations:
Init, Isempty, Isfull [15 Marks]

Q3.Viva [5 Marks]

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Q1. Write a python program to check whether an element exists within a tuple.

[15 Marks]

Q2. Create Stack – and perform - [15 Marks]

1. Implementation of an algorithm that reverses string of characters using stack and checks whether a string is a palindrome.

Q3.Viva [5 Marks]

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Q1. Write a python program to do iteration over sets. [15 Marks]

Q2 Create Circular [15 Marks]

1. Implementation of circular queue

Q3.Viva [5 Marks]

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Q1. Write a python program to add and remove operation on set. [15Marks]

Q2. Create Searching Algorithms - Implementation of searching algorithms to search an

element using: Binary Search [15 Marks]

Q3.Viva [5 Marks]

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Q1. Write a python program which reverse given string and displays both original and reversed string. (Don't use built-in function) [15 Marks]

Q2. Sorting Algorithms - Implementation of sorting algorithms: [15 Marks]

Insertion Sort.

Q3. Viva [5 Marks]

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Q.1 Write a python program to accept and convert string in uppercase or vice versa. [15 Marks]

Q2. Dynamic implementation of Singly Linked List to perform following [15 Marks]

operations: Create, Insert, Delete, Display, Search

Q3.Viva [5marks]

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Q.1 Write a python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x:x*x).

Expected Output: {1:1,2:4,3:9…..and so on} [15 Marks]

Q2. Create Linked List Applications - Merge two sorted lists. [15 Marks]

Q3.Viva [5 Marks]

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Q1. Write a python program to create an array of integers and display the array elements. Access individual elements through indexes. [15 Marks]

Q2. Create Stack - Dynamic implementation of Stack to perform following Operations: Init, Isempty, Isfull [15 Marks]

Q3. Viva [5 Marks]

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Q.1 Write a python program to get the number of occurances of specified elements in an array.

[15

Marks] Q2. Create searching algorithms (Implementation of searching algorithms) to

search an element using: Linear Search. [15 Marks]

Q3. Viva [5 Marks]

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Q1. Write a python program to reverse the order of the items in the array. [15 Marks]

Q2. Create searching algorithms (Implementation of searching algorithms) to search an element using: Binary Search. [15 Marks]

Q3. Viva [5 Marks]

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Q1. Write a python program to calculate the cube of all numbers from 1 to n.

[15 marks]

Q2. Create Linear Queue - Dynamic implementation of linear Queue to perform following operations: Init, IsEmpty, IsFull. [15 Marks]

Q3. Viva [5 Marks]

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Q.1Write a python function to display the sum of all the elements in a list.

[15 Marks]

Q2. Sorting Algorithms - Implementation of sorting algorithms: [15 Marks]
Merge Sort.

Q3. Viva [5 Marks]

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Q1. Write a python function to calculate the factorial of a number. The function accept the number as an argument. [15 marks]

Q2. Create Searching Algorithms - Implementation of searching algorithms to search an element using: Binary Search. [15 Marks]

Q3. Viva [5 Marks]

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Q1. Write a python function to check whether a number falls within a given range

[15 marks]

Q2. Sorting Algorithms - Implementation of sorting algorithms: [15 Marks]
Quick Sort.

Q3. Viva [5 Marks]

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Q1. Write a python function that takes a list and returns a new list with distinct elements from the first list.

[15 Marks]

Q2. Sorting Algorithms - Implementation of sorting algorithms: [15 Marks]

Quick Sort.

Q3. Viva [5 Marks]

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Q.1Write a python program to accept and convert string in uppercase or vice versa.
[15 Marks]

Q2. Sorting Algorithms - Implementation of sorting algorithms: [15 Marks]
Bubble Sort.

Q3. Viva [5 Marks]

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Q1. Write a python program to find square of given number using list comprehension. [15 Marks]

Q2. Sorting Algorithms - Implementation of sorting algorithms: [15 Marks]
Insertion Sort.

Q3. Viva [5 Marks]

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Q1. Write a python program to calculate the cube of all numbers from 1 to n. [15 Marks]

2. Infix to postfix conversion. Evaluation of postfix expression. [15 Marks]

Q3. Viva [5 Marks]